Universal Control Room Interface (UCRI)

Implementation recommendations and notes on their use, part control center – control center

Version 2.0

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1 Introduction

This document contains important explanations, notes, and implementation conditions for the manufacturer-independent interface between operations control systems from a wide range of manufacturers.

The control center landscape, especially in Germany, is currently characterized by technical limitations in cooperation between control centers. This situation continues in the management of operational resources: different manufacturer-specific approaches are pursued and proprietary technologies are used.

This results in media breaks and security risks that have far-reaching and adverse effects. Examples of this include disproportionately high costs and incompatibility in data communication. From the authors' point of view, this is a situation that is hardly tolerable and difficult to justify to society! Another consequence is the costs to society, which are being driven up unnecessarily without any noticeable technical progress having been made in recent years.

Against this background, the "UCRI Expert Forum" (formerly "Expert Forum Universal Control Room Interface - EFUL"). The aim of the expert forum is to define a universal control room interface for stationary and mobile voice and data applications with the following scope:

- Developing and maintaining the definition for the Universal Control Room Interface "UCRI" (Universal Control Room Interface),
- Publishing the results for non-discriminatory use by providers and consumers,
- Continuously ensuring the quality of the results.

The expert forum includes the major manufacturers represented on the German market as well as representatives of user associations and consulting firms.

The basic idea of a universal control room interface includes:

- Technical and cross-manufacturer mapping of basic processes for more efficient implementation,
- Overcoming media breaks between control centers with the help of a cross-manufacturer interface between dispatch systems from a wide range of manufacturers,
- Simplifying communication between the operational resources and the operational control systems of different manufacturers to create a common, uniform basis for data communication.
- Ensuring the security level of networked data communication between state-of-the-art operations control systems,
- Preserving the diverse supplier landscape through open standards to permanently strengthen the resilience of critical control center infrastructure through heterogeneous operations control system architectures from different manufacturers.

The present implementation recommendations and the associated interface specification in version 2.0 are a further development of version 1.0. From the perspective of the expert forum, this is not final. Therefore, work is already underway on the next version with additional functional features. A procedure has been established for publication and further development, which is explained in more detail in Chapter 5, Publication and Updating.

1.1 Technical Documentation

Technical details on specific implementation by a manufacturer are published on GitHub.

The interface specification for UCRI, control center section - control center = "UCRI L2L" can be found at the following link

https://github.com/PMeV-EFUL/ucri2/

Subfolder for main technical specifications and app documentation: https://github.com/PMeV-EFUL/ucri2/tree/main/docs

An introductory document for the apps can be found at: https://github.com/PMeV-EFUL/ucri2/blob/main/docs/apps/UCRI2%20Apps%20im%20%C3%9Cberblick.p df

The machine-readable parts of the specification can be found at: https://github.com/PMeV-EFUL/ucri2/tree/main/api/crm/2.0.0 (transport layer)

https://github.com/PMeV-EFUL/ucri2/tree/main/apps (UCRI2 apps)

2 Initial situation and objectives of the universal control center interface

2.1 Initial situation

There are currently more than 200 BOS control centers on the German market, as well as a considerable number of private control centers (energy suppliers, traffic control centers, plant fire departments, security control centers, associations of statutory health insurance physicians, and many more).

The requirements for efficient cooperation between these control centers are increasing due to the need for mutual cooperation during missions, the fastest possible provision of resources, relief during excessive emergency call volumes, or even technical failures. In addition, there is growing demand for digital cooperation with other control centers operated by KRITIS operators, utilities, etc.

The technical capabilities for the seamless exchange of mission-critical data are fundamentally in place. Nevertheless, missions and mission information are still often exchanged by telephone, email, and fax.

The reasons for the low level of digital networking between the operations control systems and operational resources are the proprietary interface solutions currently used by manufacturers and the different data structures. Furthermore, there are different master data sets, which are based less on the technology used and more on the organizational differences in the process logic of the control centers. One example of this is keyword catalogs, which differ in almost every control center.

This is no longer appropriate and requires disproportionately high expenditure, which is reflected in the costs of setting up another proprietary interface solution. In addition, such a solution is very prone to errors due to language translation errors. It requires constant, time-consuming coordination and adjustment between the communication partners and often poses risks to the security architecture. The possibilities for documentation and traceability are limited or, in some cases, non-existent.

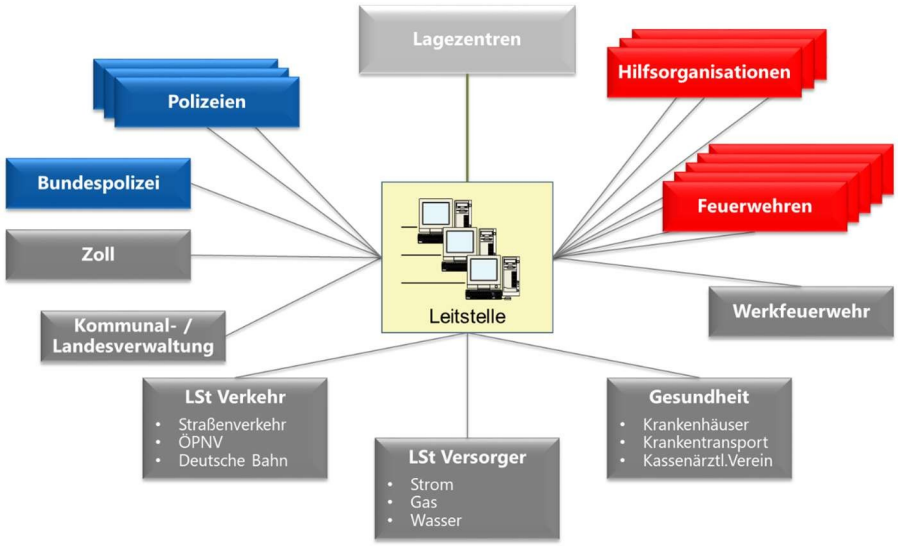


Figure 1: Interface partners of a control center

2.2 Motivation and goals for a universal control center interface

Every second counts! In an emergency, control centers and their systems must therefore be able to communicate with each other digitally – regardless of the manufacturer of the operations control system used or the master data used in it.

While transferring a recorded emergency call from one control center to another by telephone takes around two minutes on average, the digital transmission of the report is completed in milliseconds. This saves valuable seconds for the person seeking help.

It is irrelevant whether it is BOS (non-police and police), public transport, EVU, or other control centers relevant to hazard prevention that need to exchange data with each other. It must be possible to transfer data about a mission or a situation non-verbally to partners who need to be involved in or informed about the event. Control center employees must be relieved of unnecessary and error-prone work steps such as listening, understanding, taking notes, and ultimately entering data into IT systems.

The aim is to save effort and minimize errors. This is particularly important in light of the growing shortage of skilled workers.

Due to different areas of responsibility (federal government, states, municipalities, private sector), there are different terms and procedures. The goal of the universal control center interface is to connect the operations control systems and process management systems of different manufacturers and control centers in order to exchange data securely and with minimal effort.
In doing so, the principles of data protection and the applicable provisions of information security must be observed and the technical implementation must be as simple as possible.

The members of the expert forum have set themselves the goal of defining a universal interface that can be implemented by any manufacturer in its system or systems. Manufacturers and potential users outside the forum will also have the opportunity to use this interface definition.

2.3 Requirements for the universal control center interface ( )

A universal control center interface should support data communication between all types of BOS control centers, security control centers, and EVU control centers (control center to control center, or "L2L" for short). It should also enable manufacturer-independent communication between an operations control system and remote mobile software systems used by resources and emergency personnel. Examples of this include alarm apps on smartphones, guard alarm information systems, mobile patient documentation systems in Emergency Medical Services, command and support applications for incident commanders and other command facilities (control center to mobile, or "L2M" for short).

However, the applicability of the universal control center interface requires system support for control center processes. This results in fundamental requirements:

- The universal control center interface must be able to be used for individual control centers as well as for centralized systems or state solutions.
- The universal control center interface must be manufacturer-independent and the documentation must be transparently accessible to everyone.
- The functionality must be platform-independent.
- It must be a pure IP interface.
- The universal control center interface must also function without central services.

- The protection of confidentiality, integrity, availability, and authenticity must be ensured by state-of-the-art measures. These include the use of cryptographic methods, the zero-trust approach, data minimization, the avoidance of unnecessary cloud usage, network and system monitoring, and redundancy capability (see sections 4.5 and 4.6).
- If vulnerabilities in the interface specification are identified in the future, these must be analyzed by the UCRI expert forum. A revision must be initiated promptly.

The requirement to operate without central services was made with a view to preserving the independence of operators of dispatch control systems, including control center networks. In addition, internal organizational security requirements may necessitate the isolated use of the universal control center interface, i.e., without connection to a central service.

If the universal control center interface is to be connected to a central exchange platform, the expert forum recommends using a gateway solution that does not store the content data and has no access to the actual information content. This means that the mission data is encrypted by the sending control center and can only be decrypted by the intended recipient. The gateway solution therefore only has access to the metadata, which enables it to perform the transport task. However, it has no access to the actual content of the operational information to be transmitted (end-to-end encryption).

3 Function description

3.1 User

Initially, the development focus of the universal control center interface was on authorities and organizations with security tasks, i.e., the control centers of the Fire and Rescue Service, Emergency Medical Services, and police. Early on, efforts were made to also involve the associations of statutory health insurance physicians. However, the expert forum sees a significantly larger user group (Figure 2). For this reason, control center employees are referred to generally as users in the following description.

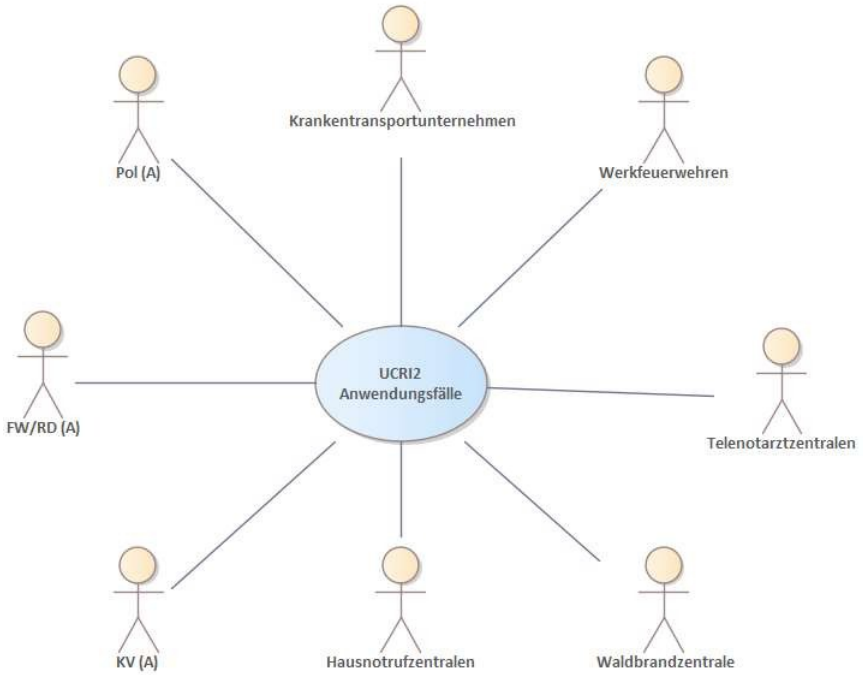


Figure 2: Potential users and stakeholders

The users of an operations control system are considered independently of their role and tasks in the respective organization. For the individual use cases, it is irrelevant whether a user in the control center assumes the role of call taker, operations coordinator/dispatcher, or another role.

3.2 e use cases considered

During the development of UCRI 2.0, specific use cases were identified and described in which interoperable communication between control centers is technically necessary. These use cases form the basis for the functional requirements of UCRI and define which information is transmitted and how the feedback is standardized technically.

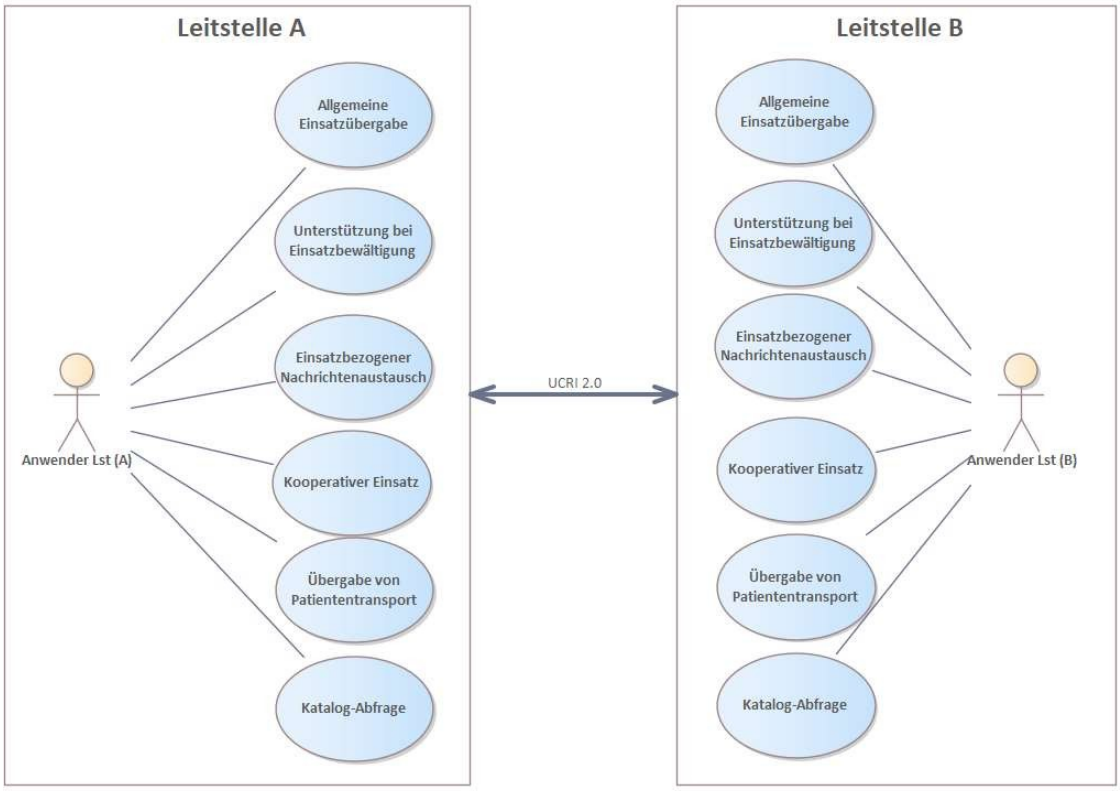


Figure 3: Use cases of UCRI

As can be seen in Figure 3, a *user in "control center A"* uses the use cases shown via the operations control system they use, and another *user in "control center B"* uses the identical use cases in the implementation of their operations control system.

The operations control systems are connected via UCRI. To this end, both operations control systems implement the "UCRI L2L" protocol. The connection is symmetrical, which means that both control centers have access to the same protocol functions. The connection is bilateral, i.e., only two control centers are involved in the communication.

Within the framework of cooperation between two control centers, there are various interactions between the people working there and between their respective operations control systems. Not all interactions are technically supported or must necessarily be mapped by the operations control system itself. In certain cases, such as the coordination of a cooperative mission, direct communication between dispatchers (e.g., by telephone) may be useful.

The structured transmission of mission-related data via UCRI 2.0 represents a significant improvement in many situations, particularly in terms of speed, data consistency, and error prevention.

The documented use cases, such as the handover of patient data, the request for resources, the exchange of mission-related messages, or the transmission of structured transport orders, each describe a clearly defined technical process that can be carried out between the mission control systems of different organizations with the support of a system. These use cases thus represent the functional objectives of UCRI 2.0 and form the basis for system-supported cooperation between control centers.

Figure 3 provides an overview of all cross-cutting use cases defined in UCRI 2.0. Each use case represents a clearly defined communication situation between control centers in which structured data is exchanged in a standardized manner via UCRI. The figure classifies the use cases according to typical communication directions and communication content and visualizes their functional relationships. Some use cases are explained in more detail below, or the use case is defined more specifically.

1. **Transfer of operations**

   Transfer of a mission from one control center to another, in particular for the transfer of responsibilities. This use case is further subdivided into:

   - "Simple handover": No personal data other than the communicator is transferred.
   - "Transfer of mission data": A mission, including structured patient data, is transferred, e.g., between emergency dispatch centers. This use case can be further detailed as "de-escalation of a non-emergency" (transfer of a mission from the 112 emergency call center to a 116117 control center of the Association of Statutory Health Insurance Physicians) or, in the opposite direction, as "emergency escalation."

   The transfer of operations also includes transfers to and from police control centers.

2. **Support in managing operations**

   Requesting resources (e.g., ambulances) from another control center to assist in managing the operation.

3. **Operation-related message exchange**

   Exchange of free text messages related to the operation for dynamic information transfer between control centers. For better structuring and clear specification, this use case is detailed in
   "Exchange of mission information," "Advance information," and "Information on the mission"
   "Third-party vehicle."

4. **Cooperative mission**

   Special form of operational support through the participation of different emergency services in a mission and the associated exchange of operational information.

5. **Patient transport handover**

   Structured commissioning of transport for one or more persons with transfer of relevant patient data.

6. **Catalog query**

   Enables a control center to query catalogs from another control center (types of resources, keywords) in a structured manner.

In this version of the interface definition, the use cases are presented in a compact f o r m . The focus is on the functionality supported by the UCRI. Therefore, the standard procedures are described.
Intended alternative processes and the handling of disruptions in the processes are explained in the protocol descriptions.

3.2.1 Operation handover

Description	
Name	General handover (simple handover (without persons) or handover with patient data persons)
Actor:	User in control center A (transferring control center) User in control center B (responsible/cooperating control center) – accepts or rejects
Benefit:	The user of the operations control system wants to transfer a mission that they has created, e.g., while processing an incoming emergency call, and for which his control center A is not responsible, with or without structured personal data, without media discontinuity and with minimal delay, to control center B, which is responsible for the mission. The result of successful execution is that the mission has been transmitted to the operations control system of control center B and accepted by control center B; the mission is processed there and control center A has received the acceptance confirmation. (Optionally, control center B can transmit an operation end message.)
Trigger:	The user in control center A (e.g., call taker) determines that the mission falls under the jurisdiction of another control center B.

Prerequisites and results	
Prerequisite 1:	The user in control center A has created a mission (e.g., to process an emergency call).
Prerequisite 2:	The basic call-out data has been entered into the control center A's call management system to at least the extent that a transfer is possible: with or without structured personal data; the call-out location has been entered. The technical connection between control center A and control center B exists.
Result 1:	The Mission Data recorded in control center A has been transferred to the task management system of control center B; control center B has accepted or rejected the transfer and acknowledged this.
Result 2:	Further processing of the mission in control center A (e.g., completion or follow-up documentation) is carried out in accordance with local specifications.
Result 3:	If successful (acceptance by B), the mission is created/accepted in control center B and displayed to users for processing; from this point on, all further information and messages are clearly assigned to this mission.
Note:	After completion, control center B can send an end-of-mission message to control center A.

Processes	
Success (technical signaling only)	Normal process
B1. Operation control system sends incident ⟶ UCRI transport can provide a technical acknowledgment of receipt. B2. Important: This acknowledgment belongs to the transport layer; the app does not define its own message for this, and the transfer is therefore not complete. Only when the receiving control center accepts or rejects the message is the transfer completed with the sending of an acknowledgment.	
Success (with technical acceptance)	Alternative procedure
A1. The user selects the mission to be transferred and the receiving control center B. A2. Dispatch system A sends the mission to dispatch system B with the following minimum content: sharedIncidentID (), sentByDispatcherAt (), missionLocation (). A3. Dispatch system B signals the newly received mission to the users in control center B. A4. Dispatch system B creates the mission. A5. The user checks and decides whether to accept or reject the assignment. The transfer is only considered complete once this confirmation has been received. A6 Mission system B confirms mission system A's acceptance or rejection of the mission with a timestamp and, if applicable, a reason (technical acceptance). A7. The A mission system signals to users in control center A that the mission has been accepted (technical acceptance). Whether the mission in A remains closed or open for subsequent documentation is not regulated in UCRI 2.0. A8. Optional: After the mission has been completed in mission control system B, the latter sends the completion message to mission control system A (including sharedIncidentID, completedAt, status, additionalInformation).	
Special cases	Technical rejection
Technical rejection in control center B: Procedure as described above up to step A5. Subsequently, signal "rejected" including optional reason in control center A. Control center A leaves the mission as is or reactivates it.	
Time out	Exception
1. UCRI 2.0 does not define any app message deadlines in the application layer. Handling within the transport layer.	
failure	Exception
1. Errors/undeliverability are handled via the transport layer. There is no app-specific error message within the application layer.	

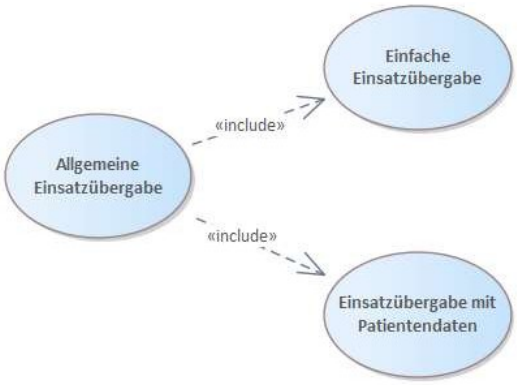


Figure 4: Transfer of duties without persons – structure of the use case "Transfer of duties without persons"

Figure 4 shows the structural relationship between the higher-level use case "General handover" and two specific variants: simple handover without personal data and handover with patient data. Both special cases are implemented as independent use cases and are integrated depending on the content and context of the handover. The general handover thus comprises the basic logic, while the specific variants represent the respective data-specific requirements.

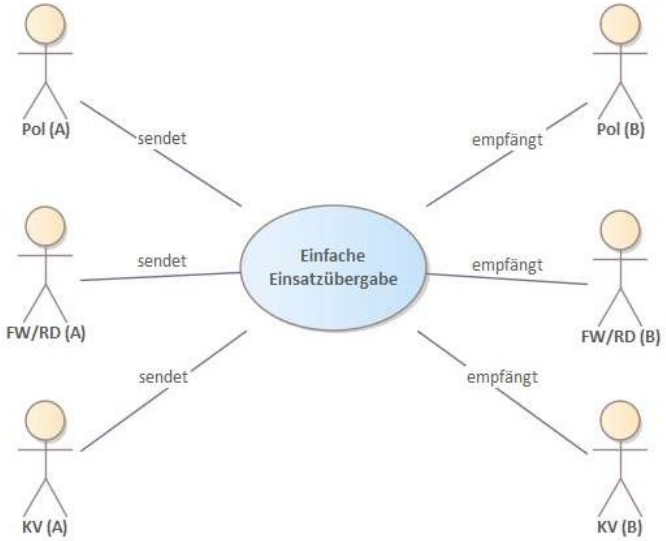


Figure 5: Handover without person – actors and communication directions

Figure 5 shows the possible sender and receiver roles in the use case of simple task handover. It shows examples of organizations such as the police, Fire and Rescue Service, and Emergency Medical Services, each in role A (sending) and B (receiving).

Figure 5 clearly shows that the use case can be applied across organizations and supports the handover of missions between different control centers, regardless of the responsible authority. The basis is always a structured handover without personal data.

The interface enables the structured transfer of Mission Data between two control centers. It supports both the sending and receiving of operational information and the feedback regarding acceptance or rejection by the receiving party. The content structure (attributes) and format of the transmitted messages are defined in the UCRI protocol for this use case.

The technical control of the transfer (e.g., which missions are transferred, how users are notified, or whether a mission remains open or is completed in the sending system) is the responsibility of the respective operations control systems. The interface only regulates the data flow, not the behavior of the systems or users.

Explanations:

1. Distinction between 'mission' and 'emergency call'

   The application only concerns the transfer of Mission Data. Any existing voice connection or emergency call channel between the control centers is not transferred or controlled via this interface.

2. Confirmation of acceptance

   The receiving control center acknowledges receipt of the mission to the sending control center. This feedback can be in the form of technical or professional acceptance, depending on the respective configuration. Feedback on the further course of the mission (e.g., dispatching, measures, completion) is not part of this use case.

3.2.1.1 Sub-case: Patient transfer – Transfer 112 ->116117

Description	
Name	Patient handover – handover 112 -> 116117
Actor	User in control center A (transferring control center) User in control center B (responsible control center) – accepts or rejects
Benefit:	The user at the 112 control center transfers a case with structured patient data to the 116117 control center if it turns out that there is no acute emergency (de-escalation). The transfer serves to hand over responsibility to the competent authority and is carried out without media discontinuity in the standardized UCRI format. Special feature of this use case: Address information describes the patient's address (master data) and not the location of the emergency. The patient can be reached by telephone; the data is transmitted in a structured manner (including personal details, insurance information, initial assessment, infection status). The transfer is considered complete as soon as the 116117 control center sends a confirmation (acceptance/rejection); optionally, a mission completion message can then be transmitted. Technical receipts are sent via the transport layer if required and are not part of the app messages.
Trigger:	The user at the 112 control center (e.g., call taker/dispatcher) determines that the case is not an emergency and should be transferred to 116117; the patient data is available, and the patient can be reached by phone.

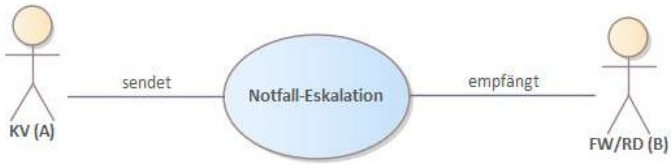


Figure 6: Special cases of transfer with patient data

Figure 6 shows an emergency escalation in the context of the use case "Patient transfer – transfer 112 -> 116117." In this scenario, the health insurance control center (KV) determines during a patient contact that this is an acute medical emergency that can no longer be treated within the scope of standard care (116117).

The structured case with patient data is therefore transferred via the interface to the responsible rescue control center (Fire and Rescue Service/Emergency Medical Services). The transfer takes place without media discontinuity and, in addition to mission data and address data, also contains structured information on the patient's state of health, PZC code, or infection status. The receiving control center decides on acceptance on this basis and, if the response is positive, takes over further care of the patient.

This is a special case of the use case "Operation handover with patient data." The basic procedure is therefore identical, and further information can be found in the parent use case.

The other processes, preconditions, and results are identical to the general handover of operations and are therefore not mentioned again for this "sub-use case."

3.2.1.2 Sub-case: Handover of police missions

Description	
Name:	Mission handover – police operation
Actor:	User in control center A (transferring police control center) User in control center B (responsible police control center) – accepts or rejects
Benefit:	The user in control center A wants to transfer a police mission to another, responsible police control center, e.g., in the event of a change of jurisdiction between federal states, regions, or organizational units. The transfer is standardized and structured in accordance with the UCRI protocol. Only information relevant to the operation is transferred, without personal data, in particular regarding the situation description, type of operation, urgency, location, and prioritized measures. The transfer is considered complete as soon as control center B accepts the mission (technical acknowledgment). Optionally, an end-of-mission message can be sent afterwards.
Trigger:	A user in control center A determines that the mission is outside their area of responsibility, e.g., in the case of cross-state operations or changing regional responsibilities, and transfers the mission to the responsible police control center (control center B).



Figure 7: Transfer of police operations

Figure 7 shows the use case of a structured transfer of operations between two police control centers. A sending control center (Pol A) transfers a police mission to a receiving control center (Pol B) if, for example, there is a change in jurisdiction, such as in cross-state missions or organizational transitions.

Communication is standardized via the UCRI interface and includes structured mission data such as location, type of mission, and urgency, but no personal data. The receiving control center decides whether to accept or reject the mission and sends a corresponding response.

The interface supports the standardized transfer of police missions between two control centers. It enables the reliable sending and receiving of relevant mission data and feedback from the receiving party as to whether the mission is accepted or rejected. The content and format of the transmitted information (e.g., mission ID, location, and mission type) are clearly defined and bindingly regulated in the UCRI protocol.

The technical control of the handover (e.g., when and under what conditions a mission is handed over, how it is displayed in the systems, or whether it is closed in the sending control center after the handover) is the responsibility of the respective operations control systems used. The interface itself only regulates the structured exchange of data between the systems.

Explanations:

1. Distinction between 'mission' and 'communication link'

   Only structured mission information is transferred. A parallel voice or radio connection between the control centers is not part of the protocol and is handled via other channels.

2. Feedback on acceptance

   The receiving control center confirms acceptance or rejection of the mission by means of structured feedback. This feedback includes the status (accepted or rejected), a timestamp, and, if necessary, a reason. No further feedback on the processing or operational implementation of the mission is provided.

The further procedures, preconditions, and results are identical to the general assignment handover and are therefore not mentioned again for this
"sub-application case."

3.2.2 Support in managing operations (request for resources)

Description	
Name:	Request for resources
Actor:	User in control center A (requesting control center) User in control center B (supporting control center)
Benefit:	The user in control center A wants to request additional ressource to handle an ongoing operation that are not available within their own organization or region. This request can be sent specifically to control center B via the interface. The result of successful execution is that suitable ressource are made available by the supporting control center B. Feedback regarding acceptance, rejection, or alternative proposals is provided in a structured manner via the interface.
Trigger:	Users in the operations control system of control center A determine that additional ressource are needed for an ongoing mission that are not available locally – for example, in special situations, capacity bottlenecks, or cross-organizational support. Control center B is to provide support.

Prerequisites and results	
Prerequisite 1:	Control center A has an active mission for which additional ressource are required.
Prerequisite 2:	Control center A can use the interface to request specific ressource from control center B. The technical requirements (connection, addressing, data formats) are met.
Result 1:	The requested resources are available to users of the operations control system at control center A for processing the mission in question.
Result 2:	While the requested ressource are being transferred to control center A, they are not available to users in control center B for their own operations.
Note:	The response from control center B may contain an acceptance, rejection, or alternative proposal. This feedback is provided in a structured manner via the protocol.

Processes	
Success (technical signaling only)	Normal process
B1. Operations control system A sends a resource request to operations control system B. The transmission is carried out via UCRI transport with technical acknowledgment of receipt. B2. Important: This technical acknowledgment belongs to the transport layer. The transfer is only considered complete once it has been accepted or rejected by operations control system B and an acknowledgment has been sent.	
Success (with technical acceptance)	Alternative procedure
A1. A user in control center A initiates a request for resources for a specific mission. A2. Dispatch system A sends the mission to dispatch system B with the following minimum content: sharedIncidentID (global, unique mission ID (UUID)), sentByDispatcherAt (timestamp of mission transfer), resourceRequestDetails (number, type, destination). A3. Dispatch system B signals receipt of the request. A4 User in control center B checks the request and decides whether to accept or reject it. A5. Operations control system B sends an acknowledgment with status (accepted/rejected), timestamp, and comment if applicable. A6 Operations control system A signals the feedback to the user.	
Special cases	Technical rejection
1. In the event of rejection (e.g., due to lack of availability), dispatch control system B sends an acknowledgment with the status rejected. A reason can be provided optionally. Control center A can withdraw or adjust the request.	
Time out	Exception
1. UCRI 2.0 does not define any app message deadlines in the application layer. Handling within the transport layer.	
Failure	Exception
1. Errors/undeliverability are handled via the transport layer. There are no app-specific error messages within the application layer.	

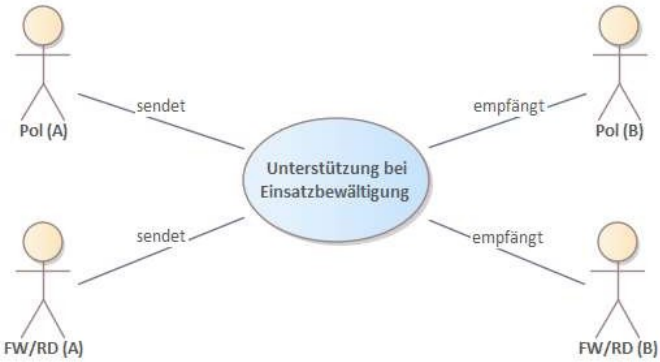


Figure 8: Request for resources

Figure 8 shows the use case of requesting resources to assist in managing an incident. A control center
(e.g., police or Fire and Rescue Service/Emergency Medical Services, represented as "A") can request a ressource from another control center ("B") if sufficient resources are not available on site.

The supporting control center receives the request and, if successful, provides suitable ressource. This process is standardized and structured via the UCRI interface. The sub-process included in the figure
"Dispatching ressource for support" describes the internal allocation or dispatching of the requested forces by the supporting control center.

The interface itself only handles communication and feedback regarding acceptance or rejection – but not operational planning or implementation.

Explanations:

1. The request for one or more resources is made in relation to a specific operation for which insufficient resources are available at control center A.
2. The request for ressource contains structured information such as the location of the operation, the alarm keyword, a description of the ressource required (specific or as a type), and other optional information. The exact structure of the request and the format of the corresponding responses (acceptance, rejection) are defined in a standardized manner in the UCRI protocol.
3. The interface can be used to request both specific ressource (e.g., vehicle identification) and generic types (e.g., "RTW"). "Command vehicle") can be requested.
4. The termination of support (e.g., after completion of the mission or return of the ressource) is implied by the status of the respective ressource and is not a separate part of the protocol.

Topics not covered

1. There is no provision for the targeted loan of resources for a specified period.
2. A return process or revocation (e.g., by control center B) is not regulated in the protocol. Manual coordination outside the interface may be necessary.
3. The flow of information regarding service times, availability, ETA, consumption, or billing of the requested ressource is not part of this use case and must be mapped within the system or via other means.

3.2.3 Operation-related message exchange ( )

Description	
Name	Operation-related message exchange
Actor	User in control center A (initiating control center) Users in control center B (cooperating/involved control center)
Benefit:	The user in control center A wants to communicate with control center B in order to effectively handle a mission without a complete mission handover. Instead, specific, mission-related information is transmitted in free text form, e.g., notes, queries, or status information. The result of the use case is a coordinated exchange between the control centers without a transfer of responsibility or resource requests.
Trigger:	Users in the operations control system of control center A recognize the need to share mission-related information with control center B, for example, in the case of parallel processing, queries, or feedback on the progress of the mission.

Prerequisites and results	
Prerequisite 1:	Control center A is handling an operation that requires feedback, information, or coordination with control center B.
Prerequisite 2:	Both control centers have an existing technical connection and support the exchange of messages via the UCRI interface.
Result 1:	The transmitted message is displayed to the receiving control center (B) in the context of the associated mission and can be processed further there.
Result 2:	Communication takes place exclusively at an informational level. No mission is handed over and no ressource is requested.
Note:	The messages are transmitted in the structured log as free text messages. A formal response is not mandatory, but can also be sent via message if necessary.

Processes	
Success (technical signaling only)	Normal process
B1. Operations control system A sends an operation-related message (notificationText) to operations control system B. The UCRI transport delivers a technical acknowledgment of receipt. B2. Important: This technical acknowledgment belongs to the transport layer. The transfer is only considered complete once it has been successfully received at the application level.	
Success (with technical acceptance)	Alternative sequence
A1. A user in control center A creates an operation-related message for an active mission. A2. Mission control system A sends the mission to mission control system B with the following minimum content: sharedIncidentID (global, unique mission ID (UUID)), sentByDispatcherAt (timestamp of the mission handover), notificationText (text content). A3. Incident management system B signals receipt of the message. A4. The message is displayed to users in control center B in context with the associated mission. Feedback is optional.	
Special cases	Technical rejection
1. The use case does not provide for formal acceptance/rejection of the message. A response can be made via a return message if necessary.	
Timeout	Exception
1. UCRI 2.0 does not define any app message deadlines in the application layer. Handling within the transport layer.	
Failure	Exception
1. Errors/undeliverability are handled via the transport layer. There are no app-specific error messages within the application layer.	

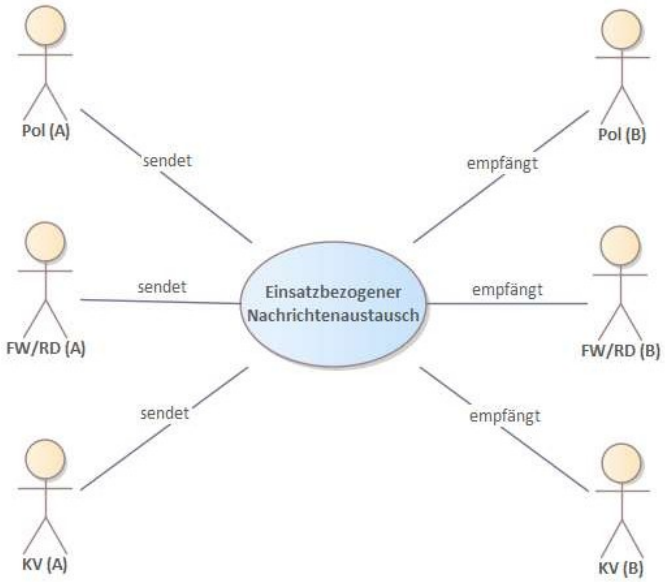


Figure 9: Mission-related message exchange

Figure 9 for the use case "Operation-related message exchange" specifically shows the exchange of information between different control center actors. Users in control center A, e.g., police, fire and Rescue Service, or association of statutory health insurance physicians, send mission-related free text messages to their respective partners in control center B. This communication takes place without a transfer of responsibility in order to convey information, queries, or status information. The use case thus supports coordinated cooperation in parallel mission processing.

3.2.4 Patient transport

Description	
Name	Patient transport
Actor	User in control center A (requesting control center) User in control center B (receiving control center)
Benefits:	The use case enables a control center to transmit a transport order for a specific patient to another control center that carries out the transport. The transmission is structured using the UCRI protocol and includes administrative details (e.g., pick-up location and destination, type of transport) as well as medically relevant information such as the patient's condition and necessary transport conditions (e.g., lying down, with medical accompaniment). The receiving control center thus receives all the information required for dispatching in a structured form and can accept or reject the order. Responsibility for organizing the transport remains with the requesting control center. The transmission of the transport order serves the purpose of inter-organizational coordination, especially in cases where transport cannot or should not be carried out by the control center originally responsible. Standardized communication ensures that all essential information can be transferred in a uniform, complete, and media-break-free manner.
Trigger:	A user in control center A recognizes that a patient transport must be carried out outside their own area of responsibility, e.g., because the destination is not within their area of responsibility or because another organizational unit is responsible for carrying it out.

Prerequisites and results	
Prerequisite 1:	Control center A has a mission that requires patient transport, e.g., for transfer to another facility or for further treatment.
Prerequisite 2:	Control center A can transmit a structured transport order to control center B via the interface. The technical requirements (connection, addressing, data formats) and minimum content requirements (including sharedIncidentId, patient, transportRequest) are met.

Prerequisites and results	
Result 1:	Control center B receives the transport order and confirms it technically and professionally. After acceptance, the order is transferred to control center B for dispatching and transport execution. Further communication regarding execution is possible.
Result 2:	If the request is rejected by control center B, control center A receives a corresponding response with the status, timestamp, and optional reason. The request is then considered not accepted and remains the responsibility of control center A.
Note:	The acceptance or rejection of the transport order is structured via an acknowledgment. Further processing (e.g., dispatching or vehicle reservation) is carried out by the system in control center B and is not part of the log.

Processes	
Success (technical signaling only)	Normal process
B1. Dispatch control system A sends a transport order (patientTransport) to dispatch control system B. The transmission is carried out via UCRI transport with technical acknowledgment of receipt. B2. Important: This technical acknowledgment belongs to the transport layer. The transfer is only considered complete once it has been accepted or rejected by operations control system B and a structured acknowledgment has been sent.	
Success (with technical acceptance)	Alternative process
A1. User in control center A initiates a patient transport order based on an existing mission. A2. Dispatch system A sends the transport request to dispatch system B with the following minimum content: sharedIncidentID (global, unique mission ID (UUID)), sentByDispatcherAt (timestamp of the mission handover), patient (patient data), transportRequestDetails (destination, transport type, etc.). A3. Dispatch system B signals receipt of the transport order. A4. The user in control center B checks the information and decides whether to accept or reject it. A5. Operation control system B sends an acknowledgment with status (accepted/rejected), timestamp, and reason, if applicable. A6. Operations control system A signals the feedback to the user.	

Processes	
Special cases	Technical rejection
1. In the event of rejection (e.g., lack of transport capacity or unclear destination), dispatch control system B sends an acknowledgment with the status rejected. A reason can be added optionally. Control center A can revise the order if necessary.	
Time out	Exception
1. UCRI 2.0 does not define any app message deadlines in the application layer. This is handled in the transport layer.	
Failure	Exception
1. Errors or transmission problems are handled via the transport layer. There is no use case-specific error definition within the application layer.	

The interface supports the standardized request and coordination of patient transport between two control centers. It enables the structured sending and receiving of transport-related information (e.g., destination, patient condition, and type of transport) as well as feedback from the receiving party as to whether the transport order has been accepted or rejected. The content and format of the transmitted data are clearly defined and bindingly regulated in the UCRI protocol for this use case.

The technical control of the process (e.g., when a patient transport is requested, what information is included, or how the status is displayed in the system) is the responsibility of the respective operations control systems used. The interface exclusively regulates the interoperable exchange of transport-related data between the systems.

Explanations:

1. Distinction between "transport order" and "communication connection"

   Only structured information relating to patient transport is transmitted. An accompanying voice or radio connection is not part of the protocol and is handled separately if necessary.

2. Feedback on acceptance

   The receiving control center acknowledges acceptance or rejection of the transport order via structured feedback. This includes the status (accepted or rejected), a timestamp, and an optional comment. Feedback on the further course of transport is not provided for in the protocol.

Topics not covered

1. There is no provision for the targeted loan of resources for a specified period.
2. A return process or revocation (e.g., by control center B) is not regulated in the protocol. Manual coordination outside the interface may be necessary.
3. The flow of information on working hours, availability, ETA, consumption, or billing for the requested ressource is not part of this use case and must be mapped within the system or via other means.

3.2.5

Description	
Name	Resource type catalog query
Actor	User in control center A (requesting control center) User in control center B (providing control center)
Benefit:	The user in control center A wants to retrieve information about the types of resources available in control center B in order, for example, to correctly select the available types or designations when requesting resources. The result of the use case is a structured list of resource types as provided by control center B. This information can be used by control center A for selection, planning, and communication.
Trigger:	The operations control system or a user in control center A needs up-to-date information about the types of resources provided by control center B, e.g., to prepare a resource request.

Prerequisites and results	
Prerequisite 1:	Control center A needs to retrieve available types of resources from control center B, e.g., to prepare a resource request or to query information about external resources.
Prerequisite 2:	Control center A can send a structured catalog query to control center B via the Interface. The technical requirements (connection, addressing, data formats) are met.
Result 1:	Control center B processes the request and sends a structured catalog of the types of resources configured in its system environment as a response. Control center A can use this information to support its decision-making.
Result 2:	If the query cannot be processed (e.g., due to technical problems), structured feedback with status information is provided. Alternative processing (e.g., manual feedback) is outside the scope of the protocol.
Note:	The catalog query is used exclusively for the structured exchange of type information. An operational resource request is not part of this use case, but is handled by the "Resource Request" use case.

Processes	
Success (technical signaling only)	Normal process
B1. Dispatch control system A sends a catalog request (resourceTypeCatalogue) to dispatch control system B. The transmission is carried out via UCRI transport with technical acknowledgment of receipt. B2. Important: This technical acknowledgment belongs to the transport layer. The transfer is only considered complete once operation control system B has responded and a structured acknowledgment has been sent.	
Success (with technical acceptance)	Alternative process
A1. The user in control center A initiates the query of the resource type catalog. A2. Operations control system A sends a query to operations control system B with the necessary identification features (e.g., sender ID, destination address, timestamp). A3. Operations control system B signals the receipt of the catalog query. A4. Dispatch system B processes the request and creates a list of available types of resources. A5. Dispatch system B sends a structured response (acknowledgment) with the status accepted or rejected, supplemented by a timestamp and, optionally, a reason. A6. Dispatch system A provides the user with the information received.	
Special cases	Technical rejection
1. If the request is rejected by control center B (e.g., due to technical restrictions or lack of configuration), a structured acknowledgment with the status rejected is sent. An optional reason can be included.	
Timeout	Exception
1. UCRI 2.0 does not define any application message deadlines. Processing takes place in the transport layer.	
Failure	Exception
1. Transmission errors are handled by the transport layer. There is no specific error definition within the application layer.	

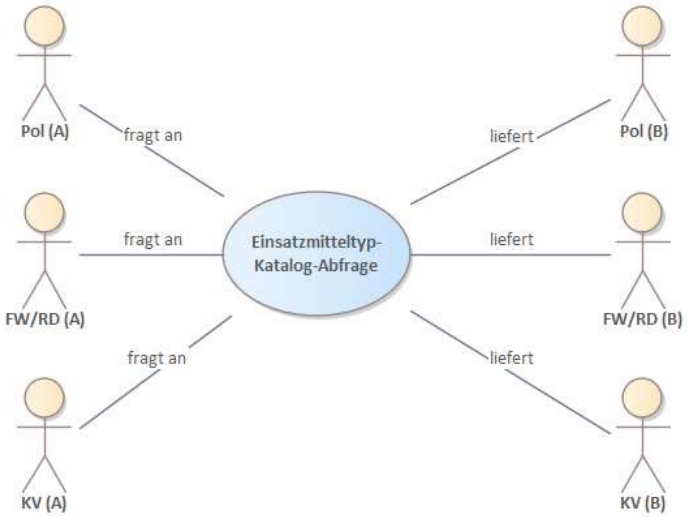


Figure 10: Resource type catalog query

Figure 10 shows the use case "Resource type catalog query," in which a control center (e.g., police, Fire and Rescue Service, or KV) sends a structured query to another control center about available resource types. The queried control center then returns the corresponding catalog. The aim is to make cataloged resource types available across organizations, e.g., for better resource planning. Communication is standardized in accordance with the UCRI protocol.

3.2.6 Keyword catalog query

Description	
Name	Keyword catalog query
Actor	User in control center A (requesting control center) User in control center B (providing control center)
Benefit:	The user in control center A wants to retrieve information on keywords or alarm terms from the classification catalog of control center B, for example to assist in the creation or further processing of missions initiated via control center B. The result of the use case is a structured catalog of available classifications (e.g., keywords, alarm categories) provided by control center B. Control center A can then use this data for its own applications, such as classification, alarms, or internal mapping of the mission.
Trigger:	A user or the mission control system in control center A needs up-to-date information about classifications (e.g., keywords) from the catalog of control center B, e.g., to correctly process a transferred mission or to align its own processes with it.

Prerequisites and results	
Prerequisite 1:	Control center A needs to retrieve available keywords or alarm classifications from control center B, e.g., to standardize incident processing or to select suitable incident keywords across systems.
Prerequisite 2:	Control center A can send a structured catalog query to control center B via the interface. The technical requirements (connection, addressing, data formats) are met.
Result 1:	Control center B processes the request and returns a structured list of keywords defined in its system environment. This can be used by control center A for systematic selection and further processing.
Result 2:	If the query cannot be processed successfully (e.g., due to technical problems or incomplete data), a structured response with error status is sent. Manual communication outside the protocol may be necessary.
Note:	The catalog query is used exclusively for the structured exchange of classified mission keywords. Operational mission processing or the transmission of a specific mission is not part of this use case.

Processes	
Success (technical signaling only)	Normal process
B1. Dispatch control system A sends a catalog query (classificationCatalogue) to dispatch control system B. The transmission takes place via UCRI transport with technical receipt confirmation. B2. Important: This technical acknowledgment belongs to the transport layer. The transfer is only considered complete once operation control system B has responded and a structured acknowledgment has been sent.	
Success (with technical acceptance)	Alternative process
A1. User in control center A initiates the keyword catalog query. A2. Operations control system A sends a structured request with the necessary Identification features (e.g., sender ID, time stamp) to dispatch system B. A3. Dispatch control system B signals the receipt of the catalog query. A4. Operations control system B processes the request and creates a list of available keywords (operation classifications). A5. Dispatch control system B sends a structured response (acknowledgment) with the status accepted or rejected, supplemented by a timestamp and, if applicable, a reason. A6. Dispatch system A makes the received information available to the user.	
Special cases	Technical rejection
1. If the request is rejected by control center B (e.g., due to technical restrictions or lack of configuration), a structured acknowledgment with the status rejected is sent. An optional reason can be included.	
Time out	Exception
1. UCRI 2.0 does not define any application message deadlines. Processing takes place in the transport layer.	
Failure	Exception
1. Transmission errors are handled by the transport layer. There is no specific error definition within the application layer.	

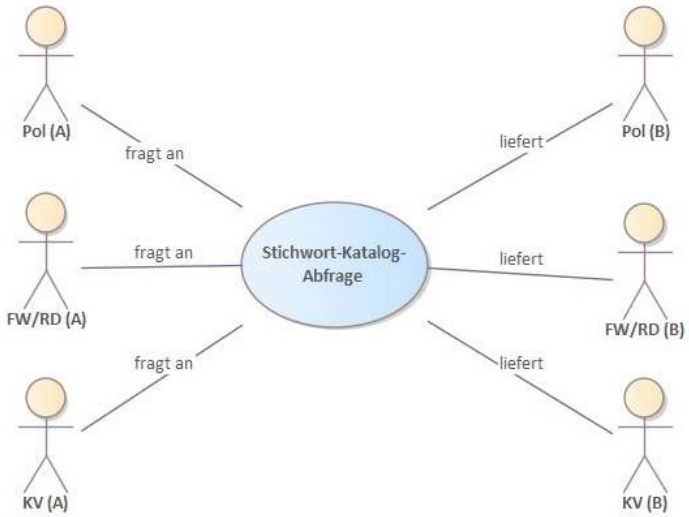


Figure 11: Keyword catalog query

Figure 11 shows the use case "keyword catalog query," in which a control center (e.g., police, Fire and Rescue Service/Emergency Medical Services, or KV) sends a structured query to another control center to retrieve stored classifications such as alarm keywords or types of operations.

The requested control center then returns its configured keyword catalog. The aim is to create a common basis for the classification and communication of missions, especially in cross-organizational cooperation.

The query is standardized via the UCRI interface and is used exclusively for structured information exchange, not for operational mission control.

4 Basic requirements for implementation and use of the Interface

This section describes the necessary prerequisites that must be met in the system (e.g., operations control system) that is to use the universal control center interface. The security and integrity of the data must be maintained.

4.1 Translation of data

Functions must be set up to translate the generic fields of the UCRI protocol to the respective local system and vice versa. The reason for this is that different manufacturers use different nomenclatures and subdivisions of information in database fields. In addition, there are currently no uniform keyword catalogs throughout Germany. This alone results in locally different configurations even for fundamentally identical products.

The fields available in the system must be mapped to the fields in the UCRI protocol. It must also be possible to split a field into several fields and merge fields (e.g., "street," "house number").

There are significant differences in the data models, particularly for location information, which must be "bridged" by this function. For example, two different logics for location data could look as follows:

- State, province, administrative district, county, city, district, street, house number, object, sub-object, location reference
- Country, district, municipality, district, location, street, street section, object, sub-object, house number

Thus, both the order and the type of sub-information for the location data differ. A generic data definition was created for the UCRI. This must be individually "translated" for each installation of a system.

4.2 Data selectivity

In general, for the interface to be usable, there is a minimum set of data fields that must always be transferred via the interface – the Minimum Set of Data (MSD).

It must be possible to define the transfer of individual fields separately for each interface partner. The transfer of data beyond the Minimum Set of Data must be configurable by the local administrator at any time. For example, it must be possible to prevent patient information from being transmitted to a police control center during emergency medical services missions. The respective configuration of data selectivity for the interface partner must enable the control center operator to implement any requirements arising from the GDPR.

4.3 Logging

Both control center systems involved in exchanging information via the UCRI must document all interactions with the UCRI in an appropriate manner. Important status information must also be displayed visibly for the user. In particular, mission-related information exchanged via the Interface must be displayed and documented in an appropriate manner.

For the purposes of case reconstruction, suitable tools must be used to enable complete documentation of the case, especially with regard to any requests for information from the public prosecutor's office.

In addition, the user of the initiating system must be notified of the success of an action in an appropriate manner, e.g., whether a transfer of responsibility has been successfully accepted by the receiving control center, or whether the requested resources can be provided. This special type of logging is essential for the fastest possible response to any negative acknowledgments.

4.4 e structure of the overall system

As a rule, data is exchanged between the individual interface partners via the Internet. Appropriate measures must therefore be taken to maintain the IT security of the control centers. Securing individual connections using encryption boxes or "normal" VPN connections and a direct connection between the systems involved are generally not options. Instead, connections between individual UCRI instances must always be encrypted, with end-to-end encryption being the goal in order to ensure data integrity.
Encryption should be viewed as part of the interface implementation. The reasons for this include:

- Maintaining security standards during transmission
- Avoiding high infrastructure costs for securing transmission routes,
- Reducing dependencies on the technical capabilities of the underlying transmission networks,
- Simplifying the technology,
- Maintaining the universal character of the interface.

With this system structure, it can be assumed that communication takes place from the secure area (control center) via a potentially insecure area (Internet). With the help of the selected mechanisms for security and encryption, it is imperative to prevent connections from being established from the insecure area to the secure area. To decouple communication, an appropriate infrastructure must be provided in the DMZ at the respective control center. BSI-compliant processes must be used for communication between the system and the infrastructure in the DMZ.

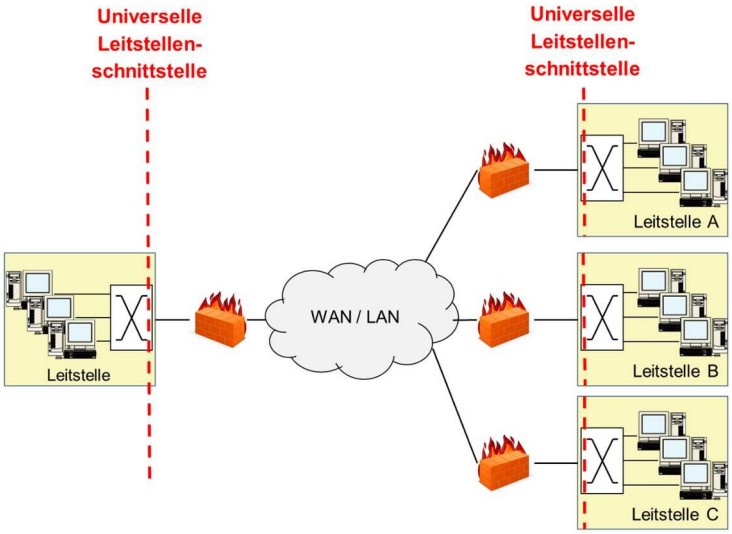


Figure 12: Overview of the overall system

4.5 's information security requirements for manufacturers

The manufacturer guarantees the safe operation of its software and associated components. It is responsible for:

- Implementing and complying with the requirements of secure software development standards,
- Implementing and complying with the contractually agreed security requirements,
- Conducting regular security audits of the organization and the software (auditing the development process and penetration testing the software),
- Providing contacts for reporting vulnerabilities and for the vulnerability management and elimination process,
- Immediate reporting of known vulnerabilities and measures to avert them,
- Timely provision of security updates for vulnerabilities and designation of a central contact person for handling and eliminating vulnerabilities,
- Conducting realistic tests of security updates in a separate environment prior to implementation.

4.6 's information security requirements for control center operators

For the holistic secure networking of control centers, minimum requirements (connection conditions) must be met by the control center operators. In addition to existing requirements, such as BSI IT baseline protection or KRITIS regulation, they form the basis for secure connection and trust between communication partners. These include:

- Development of common security objectives and (term) definitions of information security:
  - Security guidelines (strategy: What should be done? How should it be done?),
  - Protection goals (availability, integrity including authenticity, confidentiality) for data exchange between control centers,
  - Protection requirement classes (normal, high, very high) for the respective protection objectives,
  - Responsibilities.

- Identification and introduction of a procedure for implementing security requirements for the connection between control centers:
  - BSI Standard 200-2 IT-Grundschutz Methodology,
  - BSI IT Baseline Protection Compendium (including module CON.5 Development and Mission of Specialized Applications).
- Creation of target group-oriented security guidelines.
- Analysis of applicable requirements in the area of information security:
  - Minimum technical requirements, e.g.:
    - Network architecture,
    - segmentation and encapsulation,
    - network hardening,
    - Logging and monitoring,
    - Data backup,
    - Cryptographic concept.
  - Minimum organizational requirements:
    - Patch management and change management,
    - Hardware management and software management,
    - Handling of confidential information,
    - Handling and reporting of security incidents,
    - Emergency procedures,
    - Virus protection,
    - Training employees to recognize and handle security incidents,
    - Awareness raising,
    - Authorization management,
    - Dealing with external employees (service providers, outsourcing, contract data processing, etc.),
    - Documentation of security-related records.
- Auditing:
  - Verification of compliance with standards or minimum requirements (external auditors, internal auditors, mutual auditing),
  - Regulations for dealing with non-compliance.

5 Publication and updating

The aim of the UCRI expert forum is to provide operators, planners, and manufacturers of dispatch control systems and process management systems of the participating partners with a consistent, mature, manufacturer-neutral interface definition.

5.1 Publication

The aim of the expert forum and its initiation is to create an open standard that can be applied by every operator and manufacturer. The standard and the associated documents should be available for use by users in the context of procurement and requirement definitions.

The results of the expert forum will therefore be made freely available to all interested parties. They will be published on the PMeV website.

The information will be provided on two "levels":

- This document with general descriptions and regulations.
- The technical interface specification for concrete implementation by companies, which is provided in GitHub.

The concrete implementation of the interface in a system remains the responsibility of the respective manufacturer.

5.1.1 Distribution/draft versions

Only the versions made available on the PMeV website are valid and recommended for implementation. Draft versions – if distributed – are not a basis for any form of implementation, as fundamental details may change until approval.

Against this background, all members of the UCRI expert forum are also required to only distribute approved versions.

5.2 Updates

The universal control center interface standard is subject to continuous maintenance and further development in terms of applicability. It is assumed that the practical mission of UCRI 2.0 will result in a continuous need for change.

The expert forum is therefore continuing its work on the standard. Once a sufficient number of changes have been made, they will be published in a new version. The aim is to publish new versions annually.

The current version is 2.0.

For the next version, scheduled for the end of 2026, optimizations and enhancements are planned in the following areas, among others:

- Use case: Ending a mission (support for mission billing)
- Use case: Deployment information
- Use case: Appointment scheduling

From the perspective of the expert forum, these enhancements will result in a standard that surpasses most existing and currently available proprietary interfaces in terms of functionality and raises them to a new technical level.

5.3 Why is UCRI 2.0 better than UCRI versions 1.0/1.1?

The new structure of the interface and the new modularity in UCRI 2.x make it much easier to expand UCRI in the future. With UCRI 2.x, it is easier to add new interface partners, as it is no longer necessary to set up pure 1:1 connections.

We are confident that the move to UCRI 2.x will provide us with an interface description that meets the requirements of a modern and future-proof interface.

UCRI 2.x enjoys very broad support from manufacturers.

Backward compatibility with UCRI 1.x is not fundamentally excluded, but is not possible in many configurations. It is therefore necessary to clarify this with the manufacturers involved in each case.

6 List of abbreviations and glossary

Abbreviation	Explanation
BOS	Authorities and organizations with security tasks
BSI	Federal Office for Information Security
DMZ	Demilitarized zone
GDPR	General Data Protection Regulation
EDP	Electronic data processing
EFUL	Expert Forum Universal Control Center Interface
ELS	Control center operations management system
EVU	Energy supply company
IT	Information technology
KRITIS	Critical infrastructure
L2L	Control center to control center
L2M	Control center to mobile
LAN	Local Area Network
Lst	control center
MSD	Minimum Set of Data
Public transport	Public transportation
PMeV	Federal Association of Professional Mobile Communications
uc	Use Case
UCRI	Universal Control Room Interface
VPN	Virtual Private Network
WAN	Wide Area Network

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7.1 Document history

Version	Date	Status	Authors	Note
V 1.0	June 17	Approved		
V 2.0	11/03/25	In progress	K. Schneider M. Müller S. Gottwald R. Großfurtner	Complete revision and adaptation to UCRI 2.0
V 2.0	11/11/25	Approved		